

Bitcoin Blockchain Transaction Dataset for Wallet Address Profiling and Behavioral Analysis (Parquet Format)

Data Dictionary & Documentation

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1. Overview

This dataset contains **5,884,387 Bitcoin transactions** spanning blocks 744,837 to 903,456 (July 2022 to July 2025), collected and labelled to support research on CoinJoin transaction deanonymization, Bitcoin wallet behavioural profiling, and anomaly detection. Each row represents one on-chain transaction enriched with **53 structural, temporal, and heuristic features**, including a binary label (**is_coinjoin_like**) identifying transactions that exhibit CoinJoin-like mixing behaviour.

The dataset was constructed to address three open research gaps in Bitcoin privacy:

1. Taproot-upgrade clustering bypasses (P2TR / Schnorr key indistinguishability)
2. Multi-dimensional post-mix spending (CoinJoin Spending Transaction) deanonymization
3. ML class-imbalance in illicit-activity detection on transaction graphs

2. Associated Publication

This dataset serves as the primary data source for the following peer-reviewed publication:

Associated Publication Details	
Paper Title	HyGAP: A Multi-Layered Hybrid Framework for Behavioral Bitcoin Wallet Profiling and Unsupervised Anomaly Detection
Paper ID	1504
Conference	ICCUBEA 2026
Track	Blockchain And Cyber Security
Authors	Sagar Korde et al.
Status	Accept — Camera Ready Submission
Dataset Role	Primary training and evaluation dataset for wallet behavioral profiling, DBSCAN-based anomaly detection, KMeans clustering, and Structural Importance Score (SIS) computation across 950,000+ wallet addresses

The HyGAP framework chains DBSCAN-based noise isolation, KMeans clustering with a 30-day retrospective behavioural feature window, and a graph-theoretic Structural Importance Score (SIS) into a single end-to-end pipeline. Across 950,000 wallets the framework achieves a Silhouette Score of 0.68 and recovers six interpretable behavioural archetypes. The strict detection tier (top-2,847 wallets) exhibits a CoinJoin participation rate of 23.3% — 3.2x the population baseline.

3. File Format

Property	Value
File	Dataset.parquet
Format	Apache Parquet (Snappy-compressed)
Rows	5,884,387
Columns	53
Size on disk	~550 MB
Null values	None (all columns fully populated)

4. Class Distribution

Class	Count	Percentage
Normal	5,774,035	98.12%
CoinJoin-like	110,352	1.88%
Total	5,884,387	100%

5. Column Reference (Data Dictionary)

5.1 Identifiers

Column	Type	Description
txid	string	Unique 256-bit Bitcoin transaction hash (hex-encoded)
block_height	int32	Block number in which the transaction was confirmed
block_time	int64	Block timestamp as Unix epoch integer (seconds since 1970-01-01)
timestamp	datetime	Parsed UTC datetime of block confirmation

5.2 Transaction Structure

Column	Type	Description
version	int32	Bitcoin transaction version field (1 or 2)
locktime	int32	Transaction locktime field
size	int32	Raw transaction size in bytes
vsize	int32	Virtual size in vBytes (SegWit weight / 4)
weight	int32	SegWit transaction weight in Weight Units (WU)
input_count	int32	Number of transaction inputs (UTXOs being spent)
output_count	int32	Number of transaction outputs
input_addresses	list	List of input addresses (Bech32 / P2PKH / P2SH strings)
output_addresses	list	List of output addresses
input_script_types	list	Script types of each input (semicolon-joined in raw form)
output_script_types	list	Script types of each output
op_return_data	string	Raw hex payload of OP_RETURN output (empty string if absent)

5.3 Value & Fee

Column	Type	Description
total_input_value	float32	Sum of all input values (BTC)
total_output_value	float32	Sum of all output values (BTC)
fee	float32	Transaction fee in BTC (total_input_value - total_output_value)
avg_input_value	float32	Mean value per input (BTC)
avg_output_value	float32	Mean value per output (BTC)
value_difference	float32	Absolute difference between total input and output value (= fee)
fee_rate_sat_per_byte	float32	Fee in satoshis per raw byte
fee_rate_sat_per_vbyte	float32	Fee in satoshis per virtual byte (most relevant for SegWit)
value_concentration_ratio	float32	Ratio of max output value to total output value; 1.0 = single dominant output

5.4 Address & Script Counts

Column	Type	Description
input_address_count	int32	Number of distinct addresses among inputs
output_address_count	int32	Number of distinct addresses among outputs
total_addresses	int32	Total unique addresses across inputs and outputs
input_script_count	int32	Number of input scripts
output_script_count	int32	Number of output scripts
address_reuse	int32	Count of addresses that appear in both inputs and outputs

5.5 Derived Ratios

Column	Type	Description
input_output_ratio	float32	Ratio of input count to output count

5.6 Temporal Features

Column	Type	Description
hour	int32	UTC hour of confirmation (0-23)
day_of_week	int32	Day of the week (0 = Monday, 6 = Sunday)
week_of_year	int32	ISO week number (1-53)
year	int32	Year of confirmation
month_1	string	Year-month string (e.g. 2023-08) used for monthly aggregation

5.7 Script-Type Flags (Boolean)

Each flag is **True** if *any* input or output in the transaction uses that script type.

Column	Description
has_p2pk	Legacy Pay-to-Public-Key outputs present
has_p2pkh	Pay-to-Public-Key-Hash (legacy addresses starting with 1)
has_p2sh	Pay-to-Script-Hash (addresses starting with 3)
has_p2wpkh	Pay-to-Witness-Public-Key-Hash (native SegWit bc1q short)
has_p2wsh	Pay-to-Witness-Script-Hash (native SegWit bc1q long)
has_taproot	Pay-to-Taproot (Taproot bc1p - Schnorr signatures)

5.8 Transaction-Type Heuristic Flags (Boolean)

Mutually exclusive classification of transaction pattern derived from input/output structure heuristics.

Column	Description
has_coinbase	Transaction is a coinbase (block reward) transaction
has_op_return	At least one output carries an OP_RETURN (data-carrier) payload

Column	Description
<code>rbf_enabled</code>	Replace-By-Fee signalling is active (sequence number < 0xFFFFFFFF)
<code>is_self_transfer</code>	All input and output addresses are identical (self-send)
<code>is_consolidation</code>	Many inputs, few outputs (UTXO consolidation pattern)
<code>is_distribution</code>	Few inputs, many outputs (fan-out / distribution pattern)
<code>is_peer_to_peer</code>	Single input, two outputs (classic P2P payment with change)
<code>is_batch_payment</code>	Single input, many outputs (exchange batch payout pattern)
<code>is_coinjoin_like</code>	PRIMARY LABEL: exhibits CoinJoin-like mixing behaviour (multiple equal-value outputs from multiple distinct input addresses)

5.9 Metadata

Column	Type	Description
<code>sample_size</code>	int32	Total number of transactions in the block-range sample (used for stratified sampling context)

6. Labelling Methodology

A transaction is labelled `is_coinjoin_like = True` if it satisfies **all** of the following heuristic criteria:

- Multiple input addresses** — `input_address_count` ≥ 2 , indicating coordination among distinct wallet holders.
- Equal-value outputs** — at least two outputs share the same BTC value (within floating-point tolerance), consistent with standardised denomination mixing.
- Non-trivial fan-out** — `output_count` ≥ 2 , excluding simple self-transfers.
- No coinbase** — `has_coinbase` = `False`.

These criteria follow the detection logic described in the HyGAP multi-modal forensic framework (Paper ID 1504, ICCUBEA 2026).

7. Dataset Coverage

Property	Value
<code>Start date</code>	2022-07-13
<code>End date</code>	2025-07-01
<code>Start block</code>	744,837
<code>End block</code>	903,456
<code>Unique months</code>	30
<code>Taproot tx share</code>	~41% of dataset

8. Suggested Citation

If you use this dataset, please cite the accompanying paper:

Korde, S. et al. (2026). HyGAP: A Multi-Layered Hybrid Framework for Behavioral Bitcoin Wallet Profiling and Unsupervised Anomaly Detection. ICCUBEA 2026, Track: Blockchain and Cyber Security.

9. License

This dataset is released for **academic and non-commercial research use**. Redistribution with attribution is permitted. Commercial use requires explicit written consent from the authors.

10. Contact

For questions regarding dataset construction, labelling methodology, or usage:

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